1 Quick refresher on unitary transformations

Let's say we have some Hamiltonian $\hat{H}$, which we examine in a basis $\{|v_j\rangle\}$ with $j = 1, \ldots, \dim(\mathcal{H})$ and $\hat{H} \colon \mathcal{H} \to \mathcal{H}$. The matrix representation of the operator $\hat{H}$ in this basis is thus given by

$$H_{ij} = \left\langle v_i \middle| \hat{H}\, v_j \right\rangle = \langle v_i |\, \hat{H}\, |v_j\rangle\,. \tag{1}$$

Applying a unitary transformation U (with $U^\dagger U = \mathbb{1} = U U^\dagger$), i.e.

$$|v_j\rangle \to |\tilde{v}_j\rangle = U\,|v_j\rangle \qquad \& \qquad \hat{H} \to \hat{\tilde{H}} = U \hat{H} U^\dagger, \tag{2}$$

yields the matrix elements

$$\tilde{H}_{ij} = \langle \tilde{v}_i |\, \hat{\tilde{H}}\, |\tilde{v}_j\rangle = \langle v_i |\, \underbrace{U^\dagger U}_{=\mathbb{1}}\, \hat{H}\, \underbrace{U^\dagger U}_{=\mathbb{1}}\, |v_j\rangle = \langle v_i |\, \hat{H}\, |v_j\rangle = H_{ij}. \tag{3}$$

Hence the matrix representation of an operator does *not* change under a unitary transformation.